

COMPARISON BETWEEN HUMAN AND NON-HUMAN SKELETAL SYSTEM

ANA 210
DR ANDRE



Edit with WPS Office

INTRODUCTION

- ⌘ Humans are unique among primates in that they alone practice obligatory bipedalism and their skeletons show distinctive adaptations for this form of locomotion.
- ⌘ Humans' ability to regulate their body heat over long periods of heavy activity is also unique. Their large brains are highly developed organs that allow for technology, and diversity of culture and language.
- ⌘ These qualities enable humans to travel over water, in the air, and into space. Humans live permanently in almost all terrestrial parts of the planet, and occupy inhospitable environments



INTRODUCTION

- ✘ Distinguishing between human and nonhuman bone is a task that should only be undertaken by a forensic anthropologist or individuals experienced in osteology.
- ✘ These include determining whether the suspected is actually bone, and if it bone, is it human or nonhuman?
- ✘ Numerous non osseous materials such as wood, pottery, plastics, or even stones can sometimes be mistaken for fragmented human bone



INTRODUCTION



Examples of Butchered Human and Non Human Remains

Source: <https://thesebonesofmine.wordpress.com>

Distinguishing Humans from Other Mammals

- ✕ From an anatomical perspective, humans and other non – human mammals can be very similar in their skeletal components, and because humans are mammals, they possess many of the same skeletal characteristics.
- ✕ Upon gross examination, there are two main characteristics of bones that can help make the distinction between human and nonhuman bone easy and expeditious:



Maturity

- ☒ Maturity aids in differentiating small nonhuman animals that, even after reaching adulthood, have bones that are similar in size when compared with juvenile humans.
- ☒ The most common human bones to be mistaken for animal bones are the bones of infants.
- ☒ They are sufficiently different from adult and even older children bones that they can cause considerable confusion.
- ☒ Many animals have adapted to living environments that are very different to the environments we live in; therefore, they often possess distinct skeletal elements that humans do not have.
- ☒ The bones of birds are well suited to flying because they are very light.
- ☒ Additionally, birds have unique bones that are not found in the human skeleton, including the **furculum (wishbone)** and the **synsacrum (an extended sacrum)**.



Maturity

- ⊠ The presence of these bones in a skeletal assemblage is a tip off that the bones are not human.
- ⊠ Some bony elements, though distinctive can be confused with human bones when fragmented.
- ⊠
- ⊠ Pieces of sun-bleached turtle shell can mimic bits of the human skull.
- ⊠ The key to distinguishing turtle shell from human remains is to examine the cross sections of the pieces.
- ⊠ Human skull bones have an interior composed of spongy looking cancellous bone, but turtle shells do not.

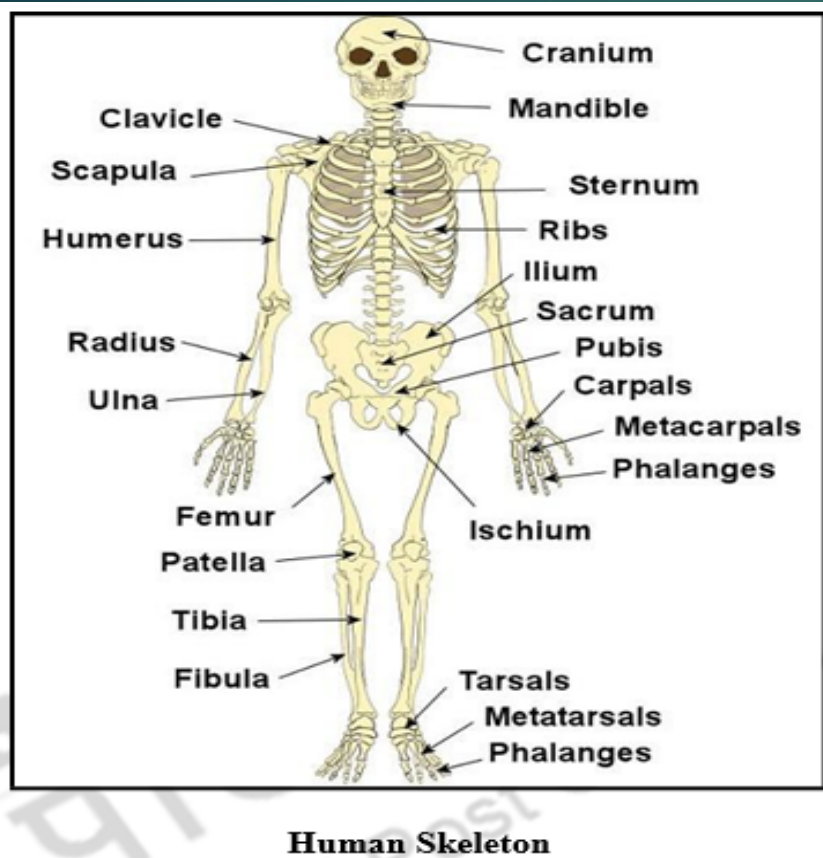


Morphology

- ✘ The second characteristic of bone that aids in the distinction between human and non human mammalian bone is morphology, or the shape of the bone.
- ✘ Because humans are bipedal they have distinct morphological features related to walking upright, which distinguish them from all quadrupeds that are adapted for four legged locomotion.
- ✘ Humans on the other hand, as bipeds, have a singular, central vertical axis of orientation that distributes all of the individual's weight through a series of bony mechanisms designed to soften the impact of bipedal locomotion.
- ✘ As a result, human crania are centrally placed on the vertical axis, the spinal column has four slight opposing curves,
- ✘ the pelvis is broad and short,
- ✘ the femora are angled,
- ✘ the tibiae have thicker proximal surfaces for greater weight bearing,
- ✘ the feet have dual arch structures, and
- ✘ the upper limbs have less pronounced musculature and a greater range of motion.



LEVELS OF BONES IDENTIFICATION



Source: <http://humansanatomy.org>

- ☒ There are generally **three levels of identification** that can be utilized to distinguish between human and animal bones.
- ☒ **Gross skeletal anatomy,**
- ☒ **Bone macrostructure, and**
- ☒ **Bone microstructure (Histology)**



1. Gross Skeletal Anatomy

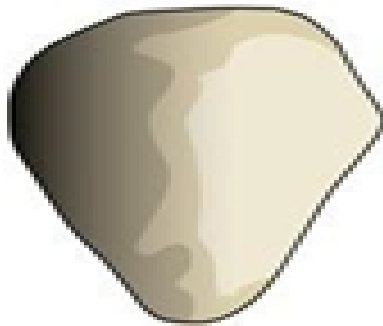
- ☒ **A. Cranium:** Cranial morphology differs dramatically between humans and animals due to the uniquely large brains that humans have compared to body mass.
- ☒ It is quite distinct because have large, rounded braincase and flat, or orthognathic, face in profile.
- ☒ Since animals (even large ones) have much smaller brains, their cranial bones are generally more curved and individually smaller.
- ☒ **Animal mandibles are often “V” shaped** in superior/inferior view and separate at the midline as opposed to the “U” shaped singular construction of the human mandible.
- ☒ Human crania are oriented on a vertical axis and the orbits are located in the front and above the nasal aperture.
- ☒ Animal crania are oriented on a horizontal axis and the orbits are located behind and lateral to the nasal aperture.
- ☒ These orientations also cause the position of the foramen magnum to be located inferiorly in humans and posterior in animals.



1. Gross Skeletal Anatomy



Long bone



Short bone



Irregular bone



Flat bone

Different Kinds of Bone

Source: <http://www.thunderboltkids.co.za>



Edit with WPS Office

1. Gross Skeletal Anatomy

- ☒ **B. Dentition** :Dentition varies greatly between humans and animals, and even between different species of animals.
- ☒ Human teeth reflect a generalized design, including a mix of slicing (incisors), puncturing (canines), and grinding (molars) teeth. They are normally more rounded than animal teeth.
- ☒ Most animal teeth reflect specialized dietary adaptations. Grazing animals have more grinding teeth with specialized ridges and carnivores have more shearing teeth with sharp ridges.
- ☒ In addition, many animals have different dental formulas compared to humans.
- ☒ Adult humans generally have a compliment of 32 teeth, eight in each quadrant.
- ☒ This includes two incisors, one canine, two premolars, and three molars (2:1:2:3).
- ☒ Although highly variable, many placental mammals exhibit a generalized dental formula that includes three incisors, one canine, four premolars, and three molars (3:1:4:3).



1. Gross Skeletal Anatomy

Differential Skeletal Anatomy of Humans and Animals: Dentition

Human	Animal
Omnivorous	Carnivorous; Herbivorous; Omnivorous
Dental formula 2:1:2:3	Basic dental formula 3:1:4:3
<u>Incisors</u> (maxillary) are larger than other mammals	Horse <u>maxillary incisors</u> are larger than human incisors
Canines small	Carnivores have large conical <u>canines</u> ; Herbivores have small or missing canines
<u>Premolars and</u> molars have low, rounded cusps divided by distinct grooves	Carnivores have sharp, pointed cheek teeth; Herbivores have broad, flat cheek teeth with parallel furrows and ridges



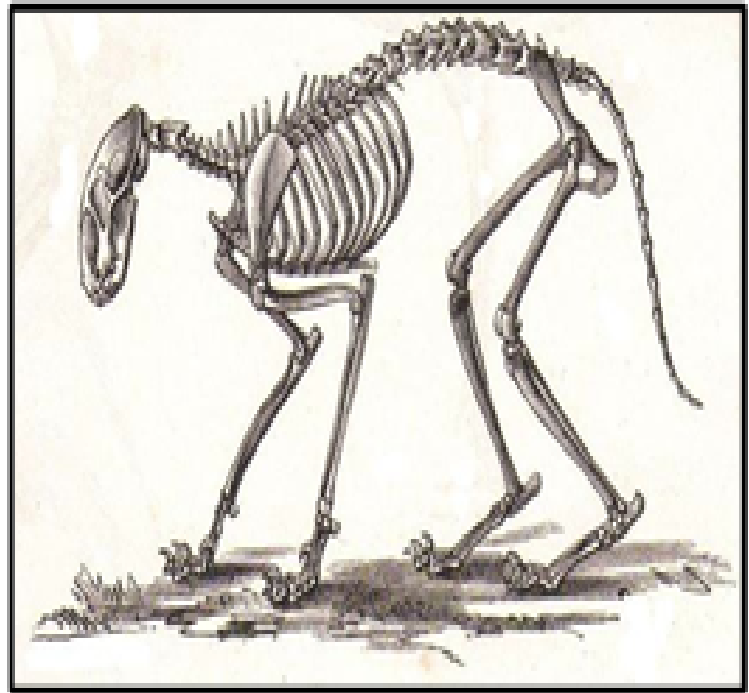
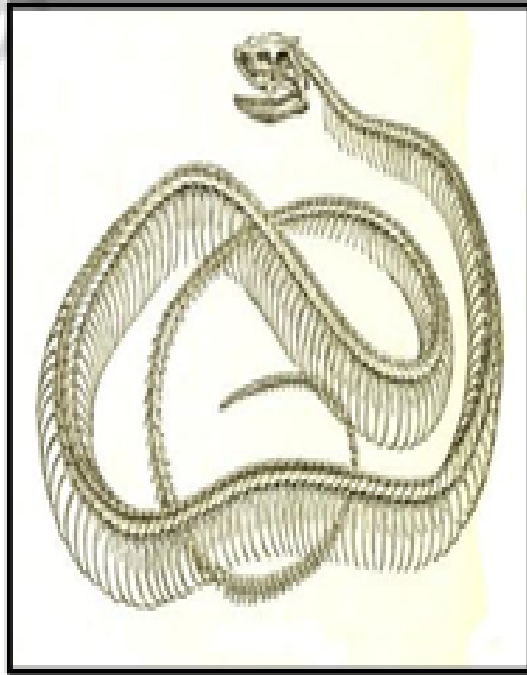
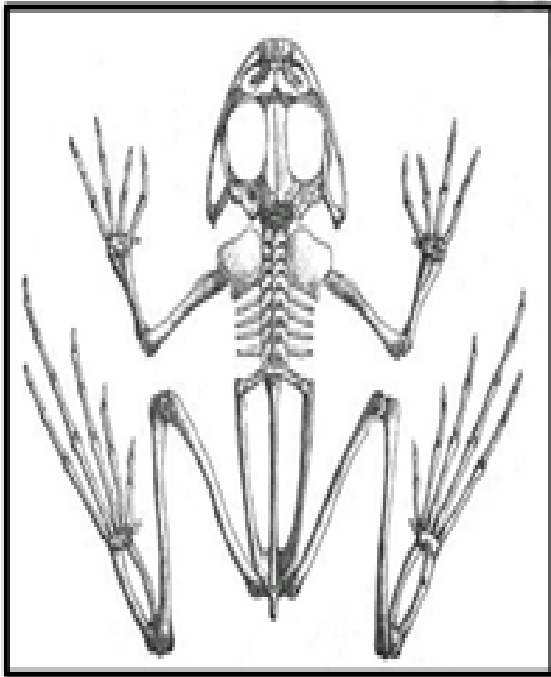
1. Gross Skeletal Anatomy

☒ C. Vertebral Column and Thorax (chest) Area

- ☒ Human and non human have about the same number of vertebrae even in giraffes have only 7 cervical or neck vertebrae but the shape of the vertebral column and of the individual vertebral bodies differ.
- ☒ The vertebral column in a typical quadruped has single gradual curve from the neck to the pelvic girdle while the human has an “s” – shaped column.
- ☒ This difference in vertebral column shape is reflected in the morphology of the vertebrae as well.
- ☒ The quadruped typically has a longer, more cylindrical vertebral body than does the human, and the vertebral bodies are more similar in length from the neck region to the pelvis.
- ☒ The thorax (chest cavity including ribs) is deep and narrow in quadrupeds and shallow and broad in humans (which brings the centre of gravity of humans closer to the vertebral column).
- ☒ Ribs straighter in quadrupeds and more curved in humans.



1. Gross Skeletal Anatomy



Skeleton of Snake, Frog, and Cat



1. Gross Skeletal Anatomy

☒ D. Clavicle

- ☒ The clavicle maintains the distance between the sternum and the scapula
- ☒ It provides support for the shoulder girdle in humans
- ☒ In some other mammals in which the forelimbs are used for manipulation, though it is vestigial or absent in many mammals and is therefore of limited use in species identification.

☒ E. Scapula

- ☒ When comparing human and other mammals, the dissimilarity in the shape of scapula can be very distinct and can quickly lead to a positive identification of human and non human remains.
- ☒ The scapula is elongated in most non human mammals, with the glenoid fossa (the point of articulation with the humerus), at the end of the long axis.
- ☒ In human the scapula is more triangular in shape, with the glenoid fossa along the most lateral surface.



1. Gross Skeletal Anatomy

☒ F. Radius and Ulna

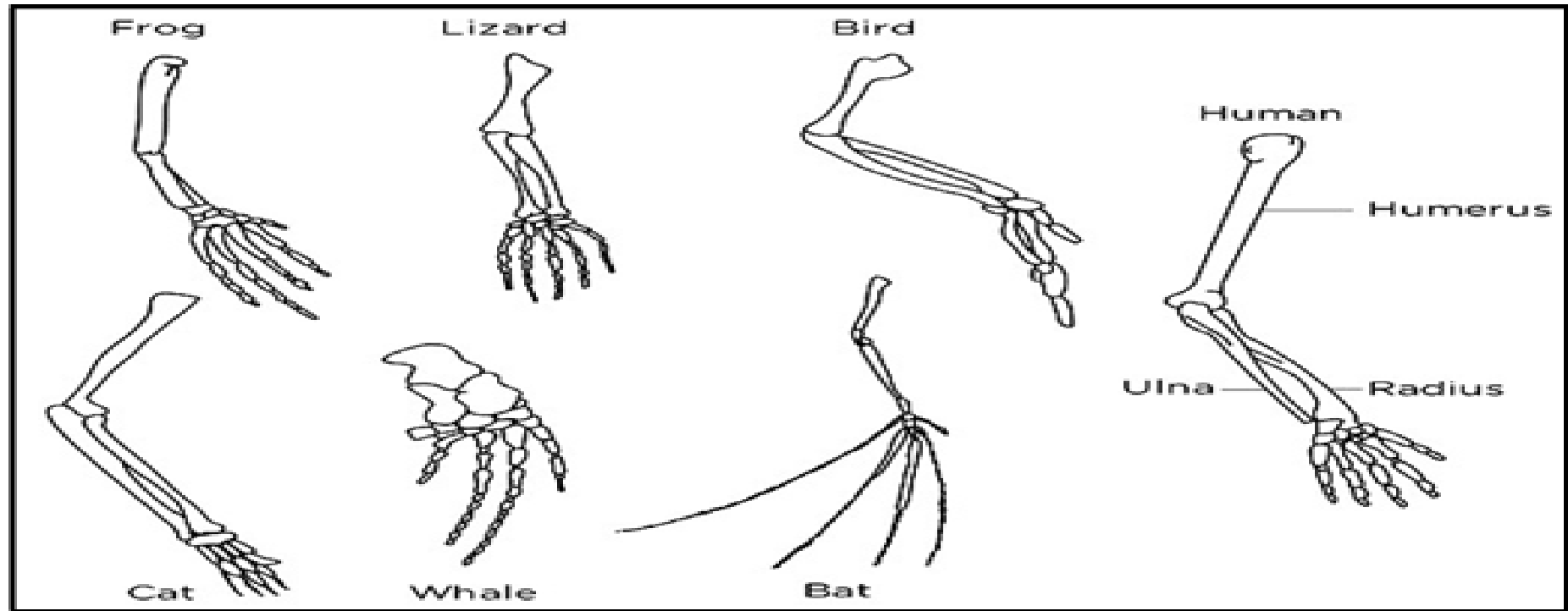
- ☒ Some larger mammal species have curved and fused radius and ulna.
- ☒ Recognition of these two bones immediately excludes humans as both the radius and ulna have straight diaphysis, and remain unfused throughout life.

☒ G. Femur / Long bone

- ☒ Much of the difference in long bone anatomy between animals and humans is the result of pattern of locomotion.
- ☒ Depending on the level of development, juvenile human long bone may have unfused separate epiphyses.
- ☒ Conversely, those of small adult non human animal will display fused epiphyses.
- ☒ As a result, non human bones can be easily differentiated from juvenile humans by examining the level of bone maturity.



1. Gross Skeletal Anatomy



Distinguish Humans from Other Mammals

Source: <http://www.thunderboltkids.co.za/>



Edit with WPS Office

1. Gross Skeletal Anatomy

⌘ H. Pelvis

- ⌘ If a small pelvis is fused in to one unit, it will be a non human pelvis because a human juvenile pelvis of a comparable size is still in multiple pieces and the two adult pelvic bones of the human do not fuse at all unless there is pathological condition.

⌘ I. Tibia and Fibula

- ⌘ In number of small animals, including mammals, the fibula is reduced in size and is fused to the shaft of the tibia.
- ⌘ In humans, the fibula does not normally fuse to the tibia unless there is pathology such as ossification of ligaments that serve to keep both bones articulated together.



1. Gross Skeletal Anatomy

I. Tibia and Fibula

- ☒ In addition, other morphological indicators that can be useful when differentiating small animals from human juvenile long bones may include non human skeletal features such as:
- ☒ the fusion of fibula and tibia, and the curvature of the long bone shaft (diaphysis).
- ☒ The long bone shaft of small mammals may be noticeably curved where it is straight in healthy juveniles

J. Foot

- ☒ Mammals belonging to the order artiodactyls (hoofed mammals that have an even number of toes on each foot – two or four) can easily distinguished from humans by the presence of metapodials.
- ☒ In these animals (for example, deer, sheep, goat) the third and fourth metacarpals and metatarsals are fused together in to one structure early in development, and the generic name for both is metapodials.



Differential Skeletal Anatomy of Humans and Animals: Post cranium

Human	Animal
Upper limbs less robust	Robust upper limbs
Radius and ulna are separate bones	Radius and ulna often fused
Large, <u>flat</u> and broad vertebral bodies with short spinous processes	Small vertebral bodies with convex/ concave surfaces and long spinous processes
<u>Sacrum with</u> 5 fused vertebrae, short and broad	Sacrum with 3 or 4 fused vertebrae, long and narrow
Pelvis is broad and short, bowl • shaped	Pelvis is long and narrow, blade • shaped
Femur is longest bone in body, <u>linea aspera</u> is singular feature	Femur is similar length to other limb bones, <u>linea aspera</u> double or plateau
Separate tibia and fibula	Tibia and fibula are often fused
<u>Foot is</u> long and narrow, weight borne on heel and toes	Foot is broad, weight borne mainly on toes



2. Bone Macrostructure

- ⌘ Animal bones have a greater density relative to size, less porous and thicker in cross section than the bones of humans.
- ⌘ In humans, humeral and femoral cortical thickness is about $\frac{1}{4}$ of the total diameter compared to about $\frac{1}{2}$ of the total diameter in animal proximal limb bones.
- ⌘ Some basic differences in animal and human bone macrostructure are:

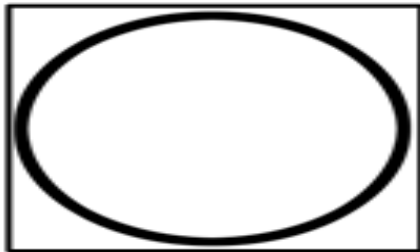


2. Bone Macrostructure

Differential Bone Macrostructure of Humans and Animals

Human	Animal
More porous cortical bone	Less porous cortical bone
1/4 thickness of diameter of long bone	1/2 thickness of diameter of long bone

**Human
animal**



Relative thickness of animal and human diaphyseal cortical bone



3. Bone Microstructure (Histology)

- ⌘ In fact, if just a small bone fragment is recovered without any morphological indicators, the only way to identify whether it is human or nonhuman would be through histological (microscopic) examination or DNA analysis.
- ⌘ In addition to the general gross morphology (shape) of the skeletal element, the external and internal textures of the bone are vital to diagnosing the bone and the species.
- ⌘ While the colour of the bone is not as important as other considerations when diagnosing species, it is important in determining taphonomic influences.
- ⌘ **Taphonomy** is defined as anything that happens to a body after death. This includes the decomposition environment and patterns (climate, water, and insects, for example, and even the temperature of the laboratory in which the remains are stored



3. Bone Microstructure (Histology)

- ✘ the post-mortem (after death) history of the remains is something one of the most important clue in solving a forensic case, and should never be dismissed when collecting evidence (including remains)
- ✘ The microscopic structure of cortical bone is often diagnostic between humans and animals, although not practical in a field setting.
- ✘ Osteons in human cortical bone are scattered and evenly spaced whereas in many animals Osteons tend to align in rows (Osteons banding) or form rectanguloid structures (plexiform bone).



Comparison of Structural and Morphological Skeletal Differences between Human and Non Human Animals

Animal	Relative Weight	Transparency	Bone Surface Morphology	Cortical Bone	Medullary Cavity	Epiphyses
Humans	Heavy	Not transparent	Well developed	Thick	Small	Distinguished
Mammals	Heavy	Not transparent	Well developed	Thick	Small	Distinguished

Birds	Light	Not transparent	Well developed	Thin	Large	Some species
Fish	Light	Transparent or semi transparent	Moderately developed	Non cancellous	Absent	Absent
Amphibians	Light	Not transparent	Poorly developed	Varies	Varies	Absent
Reptiles	Moderately heavy	Not transparent	Almost absent	Moderately thick	Reduced or absent	Varies



Summary

- ☒ There are a several ways forensic anthropologists can tell the difference between animal and human bones.
- ☒ Firstly experts have a good look at them, and compare them to their knowledge of human/animal bone shape (morphology).
- ☒ Here they are looking for the right lumps and bumps on the bone in the right places, and the general proportions of the bone.
- ☒ Big animals such as cow, sheep, and horse have much thicker, denser bone than humans.
- ☒ If tap them, they sounds much more ceramic than human bone. Expert can also look at slivers of bone under the microscope.
- ☒ Big animal bone is built differently in the body, so has a different structure.
- ☒ Animals like sheep, cows, donkeys, giraffes etc have to get up as soon as they are born to avoid predators.
- ☒ This means their bone has to be laid down very fast and has to be very tough.
- ☒ Under the microscope, it has a sort of brick like structure, where the cells are laid overlapping on top of each other like bricks in a brick wall.



Summary

- ⌘ This makes it very strong. Human bone, on the other hand, has longer to form and build strength, as when we are babies, don't have to walk immediately.
- ⌘ Humans bone cells are laid down more slowly around the blood vessels, and have a sort of concentric ring look to them, like the rings in a tree trunk.

Humans and animals have anatomical differences in skeletal composition, which distinguishes bones throughout all parts of their bodies.

Human and animal bones are distinguished by gross skeletal anatomy, bone microstructure and bone macrostructure.

Some human and animal bones are quite similar, making it difficult to identify isolated and fragmentary bones in the lab and the field.





THANKS



Edit with WPS Office